Rapport pour le comité de suivi - The socio-political lives of AI systems

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September 2023

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1 Introduction

This is the report for the first year supervising committee of the PhD thesis "The socio-political lives of AI systems : An investigation into how socio-demographic features and political attitudes guide the algorithms in social media".

Like most thesis projects, this project present a main research track, parallel research activities, and planned works for the future.

Section 2 it is possible to find an abstract of the main research track. From section 3 to section 8, the main research track is presented.

In section 9 we present a parrallel research activity. And in section 10 we present a practical timeline of this first year of work.

The planned work are separated into the different chapters under the name of "Further works".

Have a nice reading !

2 Abstract of the research

Recommender systems in social platforms attract attention in part because of their potential impact over political phenomena, such as polarization or fragmentation of online communities. These research topics are also important because of the need for understanding systemic effects in view of upcoming risk-oriented AI regulation in the EU and the US.

A common approach leverages outcomes of recommendations is to audit recommender systems. A different approach is that of explainability, seeking to render recommendation mechanisms intelligible to humans, potentially enabling both auditing and actionable design tools. This second approach is particularly challenging in the context of online systems of political opinions because of the intrinsic unobservability of opinions.

In this work we leverage multi-dimensional political opinion estimation of large online populations (along a left-right dimension but also along other political dimensions) to investigate latent spaces in representation learning computed by recommender systems.

We can see figure 1 a map of our main research track, with the numbers referring to the chapters of this document. We train a recommender based on ubiquitous collaborative filtering principles using data on content sharing on Twitter by a large population, evaluating accuracy and extracting a latent space representation leveraged by the recommender. On the other hand, we leverage multi-dimensional political opinion inference to position users in political spaces representing their opinions. We then show for the first time the relation between latent representations leveraged by a recommender system and the spatial representation of users. We show that some dimensions learned by the recommender capture ideological positions of users, bridging politics and algorithmics in our social and algorithmic system, opening a path towards political explainability of machine learning systems.

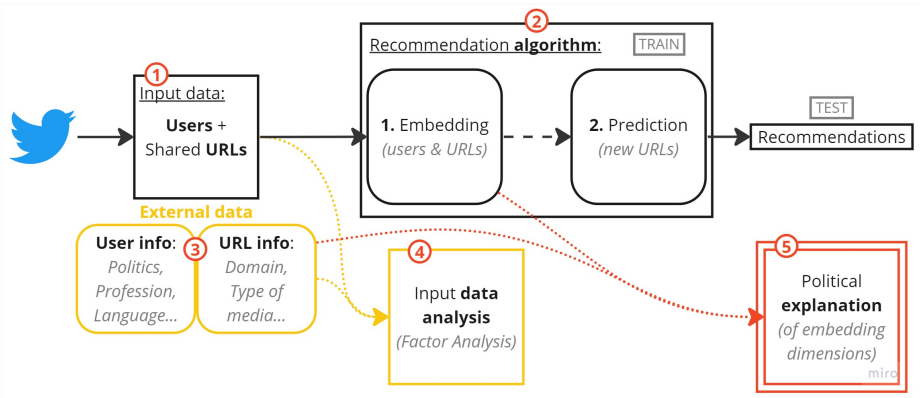


Figure 1: The plan of the research work. (1) Data collection sec 4, (2) Recommendation algorithm sec 5, (3) Socio-demographic and political data estimation sec 6, (4) Input data analysis sec 7, (5) Political explanation sec 8.

In order to test our method, we leverage other important socio-demographic features such as profession and language of the users and we show that the dimensions identified as political by our method were in fact related only to political information and not to other socio-demographic features. We then compare the factor analysis of the input data with the one of the representation space of the algorithm, and we show that, while professions and languages are the main determinant in the input data, political opinions became a significant determinant in the representations leveraged by the recommender. We argue that this not only proves our method to be reliable in explaining the mechanisms of recommendation with politics, but also opens the possibility to assess the potential responsibility of algorithms in political polarization and fragmentation.

3 State of art

Last decades have seen a growing interest in computational social sciences and in particular in social media studies [52, 53]. One reason of this interest is that social platforms play an important role in political debate in several countries, in information dissemination and consumption, and in shaping individual perceptions and political opinions [30, 76]. For this reason researchers have been particularly interested in potential "dysfunction" [79] or so-called "trouble" of the digital public space [12]. Example of such troubles being : polarization [84] (in particular affective polarization [44]), filter bubbles [67], radicalization [81], echo chambers [23] or more generally content diversity [29].

Social platforms have also multiplied the sources of information available, making recommender and information retrieval systems indispensable to navigate the large and growing set of contents [60, 13]. Recommender systems in social media have become central to platforms in their efforts to deliver satisfactory user experiences, and in driving metrics related to their underlying advertising business model [32, 45, 91]. These developments, together with the scope of the massification of social media, have raised concerns about the social and political impacts of recommender systems, and in particular in their role concerning the troubles presented before.

On the one hand several research (observational, experimental or theoretical) on both real world algorithm and theoretical models have linked recommendation algorithm with : radicalization [66], echo chambers [19, 65], biases [19, 34, 21], polarization [72] or homogenization of content [20]. On the other hand, researches has instead claimed the opposite (sometimes on the same recommender systems) showing links between recommender and content diversity [63, 34, 3, 28], or claiming that while recommenders are still bringing filter bubbles and polarization they are doing it less than users alone [64, 8]. It seems that the reason of this apparent confusion lied : (1) in the complexity of the different models and of the different fields of application, (2) in the difficulty in assessing the responsibility of the technology strongly related with the user behaviors.

To address these concerns, algorithm auditing and algorithm explainability methods have emerged as two relevant tools for evaluating, and also for understanding, how algorithms might be impacting online social systems.

Algorithm auditing involves the systematic assessment of outcomes of recommendations: i.e., what was recommended to whom. Characterizing users and items, auditing involves measuring properties of recommendations such as biases [8], diversity [75], novelty [14], or their ability to contribute to polarized states in these large online social systems [72].

Algorithm explainability, on the other hand, focuses on producing human-understandable explanations of processes deciding a recommendation, allowing an examiner to understand the steps and the factors that explain that an item is recommended over another to a given user. By understanding the underlying mechanisms, undesired outcomes can be identified and potentially avoided allowing for platform designers to make informed decisions, address concerns, and enhance the transparency and fairness of content recommendations [80].

Several algorithm explanation techniques have already been proved useful [58, 80, 90, 89].

In this thesis we aim to propose a new explanation method specific to recommender systems. In order to address the problem of the variety of recommender models and the controversial question about the role of recommender in social media troubles we introduce 2 innovations. First we propose to target and explain not the result of the recommendations, but the representative space of the algorithm. Today the vast majority of recommender systems rely on embedding representing users and items [14, 50], by targeting those embeddings we make our method model agnostic (i.e. applicable to a vast range of recommendation algorithms). Secondly we introduce for the first time vast political data as part of our explanations, this allowing social and political scientist to bring direct analysis on the explanation with no further abstraction steps.

We hope that our work will help the growing scientific community calling for better practice in coding recommendation algorithms (2006) [59]. Explicability seem essential in the technical effort to adapt accuracies and design principle[38], and resonate with the trust and accountability ethical challenges raised by the community [61].

In addition to the technical advance, our work match also well with the calls for help of the legal institutions in charge of regulating the social platforms (such as the European commission [71]) that struggle to find concrete actions [11], in part because of the lack of accessible knowledge on the subject [6].

4 Data collection

In this work we try : (1) to understand if some recommendation algorithms can infer and leverage political attitudes of users, and (2) to propose a political explanation method capable of accounting for this phenomena. Although some works has already showed relation between politics and recommendation [8], we aim to reproduce this findings on real world data with a proper measure of political attitudes and to be able to analyze directly the model representation space (beside the result of recommendation). We thus need a large scale data set of real world users for which we can estimates the political attitudes. In order to create this dataset we choose to rely on the estimation method proposed by Ramaciotti Morales et al [74] to estimate the political attitude of a large panel of users from the french Twitter sphere. This estimation will be presented section 6.1. Because the method rely on information about the members of parliament (MPs) followed by each user, we choose collect our users directly from the followers of the french MPs[1].

During this first year of research we built a **User-item dataset** where we consider a group of users (Twitter users) and a group of items (URLs), and the interactions between those users and items (users sharing URLs). The goal being to develop recommendation algorithm capable to recommend new items to each user hence capable to predict new interactions (see section 5).

Section 4.1 we present the user collection, section 4.2 the items collection.

4.1 User collection

Here is our user collection method :

- We consider the french Members of Parliament (MPs) in may 2019 : 813 out of 925 were officially on Twitter and had allowed the collect of their followers.
- We collect the 4.487.430 unique followers of the MPs.
- We remove the user following less than 3 MPs[2] and the the user with less than 25 followers[3].
- We ignore all the users following the same subset of MPs (e.g. if 3 users are following exactly the same MPs we keep only one of them)[4].

We obtain that way 368.831 users.

[1] This obviusly bring representativity issues that will be addressed.

[2] The political attitudes estimation would be too uncertain for those users

[3] This *good habit* is proposed by [9] to remove part of the fake account

[4] This choice is a simplification used by Ramaciotti hence those users are considered as equal for the attitude prediction method. We can consider the user chosen as an average representative of the group of users.

4.2 Item dataset collection

To simulate our recommendation algorithm we need data to consider as item to recommend, since views and likes cannot be widely scrapped from Twitter we choose to consider the URLs shared by the twitter users as items. We then consider the action of sharing an URL on twitter as meaningful user-item interaction.

- We randomly extract 39.950 users from the 368.831 followers of MPs (presented in section 4.1).[5]
- We collect all the tweets containing URLs shared by those users.
- We process the URLs (using ural library - tool from medialab) in order to recognise distinct internet domains (e.g. lemonde.fr, lefigaro.fr, businessinsider.de...).

We obtain after this collect 39.950 users, 426.014 domains and 23.534.803 sharings. From here we will consider the URLs as *items*, and we will call *interaction* between a user u and an item i the number of time that i has shared u (if u has never shared i we say that there is no interaction).

4.3 Further works

Since the beginning of the thesis the médialab has held several new data collections. In particular a whole new user dataset with the proper political attitudes estimation has been collected. We plan during the second year of thesis to use those data in order to get more recent and clean results.

[5] We choose to consider 39.950 users to make scale economies of resource (to collect data, train the algorithm and analyse the result) while keeping a reasonable number of user in order to obtain statistically significative results.

5 Recommendation algorithms

Once we get our data we want to train a recommendation algorithm on this data in order to test our explanation method. Modern recommendation algorithms can usually be separated in 2 phases [50] :

1. An **embedding algorithm** which map the users and the item into a multidimensional space called latent embedding.

2. A **predictor** which take as input the embedding and give for each couple (user,item) a prediction of how much the user will like the item.

A common way to tackle the embedding problem (1) is the collaborative filtering. This set of methods assume that similar user will rate similar content similarly. This allows to define a multidimensional space in which we can embed both the users and the items and in which a simple proximity measure[6] would determine the rating of each user for each item.

Collaborative filtering methods for recommendation algorithm has been introduce in the Netflix challenge in 2007 by [62] and popularized by the 2009 [51]. This popularity is in part due to 3 great advantages :

- it did not need any user or item data and rely solely on interaction data,

- it is fast and easy to implement,

- it can have fair performance on small data sets (reducing part of the so called *cold start problem*).

Because of those advantages, most of the algorithm still rely, at least in part[7], on collaborative filtering methods [14].

Because in the main part of the literature, rating predictors tend to be very simple (going from a simple dot product to a few layers neural network predictor), and because our explanation work aim to be generalizable we then choose to focus in this first proof of concept on the collaborative filtering embedding method as a simple example of embedding. In particular we focus on the Non Negative Matrix Factorization (NMF) algorithm.

5.1 Collaborative filtering model

The Non-Negative[8] Matrix Factorization (NMF) problem is a known problem in mathematics [55]. In the recommender litterature we can see many different formulations of the problem. During our work we implement 2 different formulations : (1) the mathematical matrix formulation [54] and (2) the vectorial formulation [51]. While the vectorial formulation is more complex and can give

[6] s.a. a scalar product for example

[7] many algorithm relies **also** on user based and item based assumption

[8] The non negative constraint is not systematic in all matrix factorization recommender systems. It is usually justified in problems where we seek to extract only positive influence of latent variable on interactions, and it is often said to avoid over fitting problems.

better result in litterature it is computationnaly way more expensive and don't take into account unobserved interactions as null interactions. For this reason we obtain better performances with the matrix algorithm and we hence choose to present this formalization.

Let $\mathcal{U}$ be a set of n users and $\mathcal{I}$ set of m items.

For each user and item $u, i \in \mathcal{U} \times \mathcal{I}$ we call r_{ui} the rating of the user u for the item i.

The problem of recommendation consist in predicting the future ratings from a known set of ratings.

We call $\mathcal{K} \subset (\mathcal{U} \times \mathcal{I})$ the set of the known ratings (one of the particularity being that the number of known ratings is really small compared to the unknown ratings : $|\mathcal{U}| << |\mathcal{K}| << |\mathcal{U} \times \mathcal{I}|$)

Let $R \in \mathcal{M}^{n \times m}$ be the rating matrix. The rating matrix can be defined as follow:

$$\forall (u, i) \in (\mathcal{U} \times \mathcal{I}), R_{ui} = \begin{cases} r_{ui} & \text{if } (u, i) \in \mathcal{K} \\ 0 & \text{otherwise} \end{cases}$$

Note that we take unobserved interactions as null interactions.

Let $k \in \mathbb{N}_+^*$ s.t. $k < min(n, m)$ be the number of latent dimensions.

The NMF problem is to find $P \in \mathcal{M}_+^{n \times k}$ (Users matrix) and $Q \in \mathcal{M}_+^{m \times k}$ (Items matrix) positives matrices, such that $R \approx P \cdot Q^T$.

Because the equality is proved impossible for $k < rank(R)$, we then refer to the optimization problem :

$$\min_{P \in \mathcal{M}_+^{n \times k}, Q \in \mathcal{M}_+^{m \times k}} ||R - P \cdot Q^T||_2$$

Adding regularization parameters to avoid over fitting we get :

$$\begin{aligned} \min_{P \in \mathcal{M}_+^{n \times k}, Q \in \mathcal{M}_+^{m \times k}} & 0.5 \cdot ||R - P \cdot Q^T||_2^2 \\ & + \alpha_P \cdot l_1 \cdot k \cdot ||P||_1 \\ & + \alpha_Q \cdot l_1 \cdot k \cdot ||Q||_1 \\ & + 0.5 \cdot \alpha_P \cdot (1 - l_1) \cdot k \cdot ||P||_2^2 \\ & + 0.5 \cdot \alpha_Q \cdot (1 - l_1) \cdot k \cdot ||Q||_2^2 \end{aligned}$$

Where $\alpha_P,\ \alpha_Q,\ l_1$ are parameters respectively controlling regularization on P, on Q and the norm used. $|| \cdot ||_1$ and $|| \cdot ||_2$ refer to the usual norms 1 and 2 on matrices spaces.

Figure 2 we can see a graphical representation of the Matrix Factorisation. We can identify the User matrix (P) and the Item matrix (Q) as a k-dimensional embedding of the users and the items.

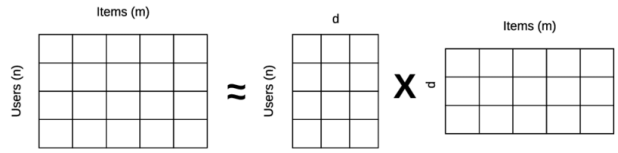


Figure 2: We can see from this graphical representation of NMF that Factorizing the Users-Items matrix (R) result in creating a User matrix (P) and an Item matrix (Q) that represent a d-dimensional embedding

Once the embedding P and Q has been computed we can predict the new ratings by simple dot product : $\hat{R} = P \cdot Q^T$.

5.2 Metrics

In order to test our performances we build from our interaction data set $\mathcal{K}$ a train set $\mathcal{K}^{\text{train}}$ with 80% of the interactions and a test set $\mathcal{K}^{\text{test}} = \mathcal{K} \setminus \mathcal{K}^{\text{train}}$.

Because in application problems the recommendation systems are used mainly to order or classify content we choose in our work to use an ordinal accuracy metric. Widely used in the literature [33] we choose the $Hits@K$ metric. This metric aim to consider the problem of recommendation as a binary classification problem, in which we seek to predict the K best contents for a given user, and to return the F1-score of this classification.

Lets call $\mathcal{T} := (\mathcal{U} \times \mathcal{I}) \setminus \mathcal{K}^{\text{train}}$, the set of all the possible interactions beside the train interactions.

For each user $u \in \mathcal{U}$, the possible items of u among $\mathcal{T}$ can be define as :

$$\mathcal{T}^i(u) = \{i \in \mathcal{I} \text{ s.t. } (u, i) \in \mathcal{T}\}$$

Let $u \in \mathcal{U}$, and $K \in \mathbb{N}_+^*$.

The set of the K best **predicted** items for u among $\mathcal{T}$ is :

$$Rec@K(u, \mathcal{T}) := \{i \in \mathcal{T}^i(u) \text{ s.t. } |\{j \in \mathcal{T}^i(u) \text{ s.t. } \hat{r}_{uj} \geq \hat{r}_{ui}\}| < K\}$$

The set of the K best **real** items for u among $\mathcal{T}$ is :

$$Pref@K(u, \mathcal{T}) := \{i \in \mathcal{T}^i(u) \text{ s.t. } |\{j \in \mathcal{T}^i(u) \text{ s.t. } r_{uj} \geq r_{ui}\}| < K\}$$

We then define the metric $Hits@K$ as the proportion of best items guessed for u :

$$Hits@K(u, \mathcal{T}) = \frac{|Rec@K(u, \mathcal{T}) \cap Pref@K(u, \mathcal{T})|}{K}$$
$$Hits@K(\mathcal{T}) = \sum_{u \in \mathcal{U}} \frac{Hits@k(u, \mathcal{T})}{|\mathcal{U}|}$$

This metric is in fact the $F1 - score$ of the classification problem of finding the best K items for each user. In this work we will use $K = 10$ mainly because of the size of our data-set. Note that because $\mathcal{T} = (\mathcal{U} \times \mathcal{I}) \setminus \mathcal{K}^{\text{train}}$ we are not only trying to predict the best contents among the test set, but among all the possible interactions. The random guess performance of this task is $Hits@10(\mathcal{T})[\text{Random Guess}] \approx 10^{-4}$.

5.3 Solver

We solve the optimization problem with the *Multiplicative Update solver*. Proposed by [54] this solver is a gradient descent with adaptive learning rate solver, which allow fast convergence and positivity of the solution.

We initialize our algorithm with a *Non-negative Double Singular Value Decomposition* method[9] as proposed by [17], to ensure faster convergence, and better results[10].

We used the algorithms proposed by [22].

We then optimize the hyper-parameters with a *Particle Swarm Optimization* (PSO) algorithm. Introduced by [48] the PSO algorithm is a non-linear non convex optimization method commonly used for hyper-parameters optimization. We train the PSO algorithm on the $Hits@10(\mathcal{T})$ metric.

5.4 Pre-processing

To improve the performance of the algorithm we clean our dataset (collected section 4).

First we eliminate from our data set all the generic URLs referring to social medias (e.g. twitter.com, facebook.com...)[11]. In fact those URLs gives us really little information about the content shared (because we consider only the domain of the shared urls), while being among the most shared ones. They then risk to have a great impact on the loss function with low improvement in accuracy. Furthermore, because those URLs are shared by a great part of the users, letting them out of the algorithm change little to the algorithmic results beside faster computation time and more accurate results. We in fact observed better accuracy scores without those URLs.

Second we recursively remove all the users having shared less than 10 different URLs and the URLs shared by less than 10 different users. We choose this treeshold (of 10) for three reasons :

1. We build our test set from 20% of the total interactions, having at least 10 different interactions for each user or item in the full data set ensure

[9] The method is inspired from SVD but ensure positivity of the factorization.

[10] By accessing better local minima.

[11] Full list : twitter.com (5.158.473 sharings), facebook.com (1.127.131), youtube.com (889.057), instagram.com (260.816), linkedin.com (91.444), amazon.com, bit.ly, fllwrs.com, unfollow.fr

us a reasonable probability to have at least one interaction for each user or item in the test set.[12]

2. We choose to evaluate our model with a Hit@10 metric (presented section 5), for this reason we try to reduce the number of users with less than 10 interactions in the train set.

3. We want to avoid the cold start problem well known for collaborative filtering methods.

Note that the removed users and items being marginal their removal had a negligible impact on the loss function and on the algorithmic result while having a huge impact on the accuracy and the computation time.

Our interactions matrix is implicit, i.e.: it do not indicate ratings but only the number of time a specific user shared a specific item. This is a common problem while developping NMF recommender systems [42]. We estimate the rating matrix R by applying a log formula to our data set :

$$r_{ui} = 1 + 0.98 \cdot log(1 + 2 \cdot s_{ui})$$

Where s_{ui} is the number of times that u shared i.

The log scale is usually recommended to go from implicit to explicit ratings [46], and the parameters (0.98 and 2) have been choose depending on the data set in order to get ratings in a 1 to 10 range. This step avoid the risk of having few interactions with an enormous impact on the loss function, while most of the other interactions are unimportant. We can notice that because log is monotonous then ordinals accuracy metrics such $Hits@10$ stay unchanged.

After the cleaning phase we obtain a rating matrix $R \in \mathcal{M}^{29.373 \times 32.639}$ with $n = 29.373$ users, $m = 32.639$ items, and $|\mathcal{K}| = 3.277.738$ ratings[13].

5.5 Results

Our trained NMF algorithm with optimized hyper-parameter by PSO perform with accuracy of $Hits@10(\mathcal{T})[\text{NMF}] = 0.35$. This result compares positively to the estimated random guess and is in line with reported performance of real-world systems [41]. We get from the algorithm an embedding with $k = 12$ latent dimensions.

[12] During the evaluation, if no item is in the test set the accuracy is considered 0. We can then expect a real accuracy higher than the measured one.

[13] This represent 0.34% of the possible interactions

6 Socio-demographic and political data estimation

Once we get our recommendation algorithm embedding we want to compare this embedding with political data. We then need to estimate the political attitudes of our users (see section 6.1). In order to test our relations toward potential confounding factors and in order to understand more profoundly our input data we also decide to estimate other important socio-demographic information : such as profession and language of the users (see section 6.2) and editorial information of the URLs (see section 6.3).

6.1 Political attitudes estimation

Estimating political opinions of individuals has been a long standing challenge for political scientists. One limitation is the difficulty in defining what a political opinion means (e.g.: is that the vote of the individual, or what he will answer if asked ?). Theoretical works from psychology introduced the notion of *attitude* as a potential solution to this problem [10, 5]. We call *political attitude* e a state of mind or a predisposition positive or negative toward a determinate politically related statement. This notion allow researcher to consider political ideology as a latent random variable that influence behavior, giving the possibility to estimate formally this value.

Some works has already leverage political attitude estimation on social media with USA centered data, using the political representative followed by each user as a meaningful trace of political attitude [15, 9]. However this kind of model was difficult to export in European case of figure because of the complexity of the European politics[14] [7].

Nonetheless, recent work by Ramaciotti Morales [74] has expanded this model showing that it is possible to infer the political opinions of certain users also in the case of more complex political spaces. Using political expert data such as the Chapel Hill Expert Survey (CHES) [47], which ranks each European party according to their political attitudes[15] towards a set of 51 different issues (e.g. European integration, special rights for minorities...), it is possible to estimate the political attitude of these users.

We hypothesize that the action of following a MP carries a certain ideological weight which is greater the more distant the ideological positions of the user and the MP are. In particular we define the probability that user i follows a MP j as :

$$P(A_{ij} = 1|\alpha_i, \beta_j, \gamma, \phi_i, \phi_j) = logit^{-1}(\alpha_i + \beta_j - \gamma|\phi_i - \phi_j|^2)$$

[14] USA political debate tend to be mono-dimensional while for most European countries this debate has 2, 3 or 4 main dimensions

[15] i.e. their positive or negative predisposition towards certain political statements or positions

Were $A_{ij} = 1$ when user i follows user j, α_i and β_j are the "activity" (tendency to follow others) and "popularity" (tendency to be followed) of users i and j, ϕ_i and ϕ_j are their latent ideological parameters, and γ is a normalization constant.

These latent ideological parameters can be approximated for a large set of users an MPs by a Correspondence Analysis (CA) of the adjacency matrix $A \in \mathcal{M}^{|Users| \times |MPs|}$ [57]. Thus allowing to project the MPs and their followers in a multidimensional ideological space. The ideological dimensions thus obtained correlate with the political analysis carried out by the experts (CHES), but lack a consistent measurement scale this preventing on the one hand to identify the political "center" and "extreme", and on the other to compare the results obtained dimension by dimension or country by country. To overcome these limitations, we apply an affine transformation to the ideological space in order to transpose it into an attitudinal space whose dimensions correspond to the attitudinal dimension of the CHES. The CHES classifies each European party according to their attitudes towards a set of 51 different political issues. The affine transformation is thus optimized in order to approximate as well as possible the position of the parties in the ideological space.

We can see fig 3 the resulting distribution of our data on the 2 main attitudinal dimensions.

Several empirical studies has shown the consistence of this estimation technique [24, 73]. In particular we can show that auto-declarations of the users in their bios are coherent with their position in the attitudinal space.

6.2 Socio-demographic estimation

After getting the political attitududes we try to estimate other characteristics from the profile of our user. In particular we tried to extract characteristics that could have strong relations with the media consumption (hence with URL sharing). We identify 2 potential counfounding factors possible to estimate from the user description : the profession and the language.

We estimated language and profession from the auto-declarative information available in the user description. In our dataset 22.087 out of 29.373 users (75%) had an available description accessible to collect. We can notice that this value is higher than for the average twitter user (around 48% of all users [26]) because the users in our dataset are more actives[16] than average Twitter users.

6.2.1 User language

We started by estimating the language of each user. We choose to rely on automated algorithm to do so. Giving the difficulty of the task on very small texts we use 3 different available python language detector : `langid.classify`, `langdetect.detect` and `cld3.get_language`. We then get the language only if at least 2 models agree on it, we instead return `False` if no language matches with another.

[16] They share URLs and follow MPs

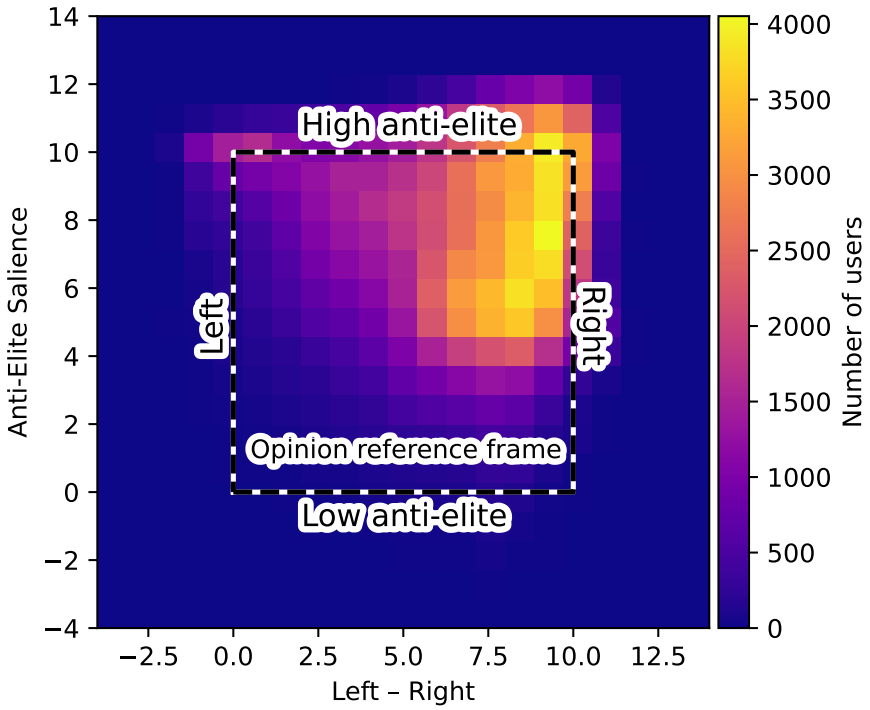


Figure 3: Spatial distribution of our user dataset on the two main attitudinal dimensions Left-Right and Anti-elite salience (Those dimensions corresponding to the CHES survey dimensions)

Language	**Number of users**
French	17956
English	2449
Spanish	132
Italian	71
German	62
Arabic	43
Portuguese	28
Catalan	24
Others	181
Unrecognized	1141

Table 1: Main languages used in the description of users on Twitter

We can see table 1 the main languages used by the users. As expected due to our collection method the great majority (92,4%) of the descriptions are French or English. However we also found a strong spanish-catalan speaking community that will reveal important in section 7.

6.2.2 Profession and socio-economic status

We then tried to estimate the professional occupation of our users. Estimating twitter users' occupation from profile information is a common task in computational social sciences, in particular in attemps to assess the socioeconomic status (SES) [31]. The easiest way to estimate users occupation is to use the auto-declarated occupation in users profile description. According to [70] 20% of active users declare their occupational status on their description. We expect this value to be a little bit higher for our dataset considering that our user tend to be more active and with higher SES than the average.

According to [37, 77] a common technique to estimate profession in user description is to automatically search for occupation words in user description, and link them to the standardized classifications like the *nomenclature des Professions et Catégories Socioprofessionnelles* [68] (CSP2020) for France, the *Standard Occupation Classification* [78] (SOC2020) for UK or USA or the *International Standard Classification of Occupations* [43] (ISCO-08). The main limitations of this technique in the goal of assessing SES (hence material, cultural and social capital) are : (1) the difficulty of distinguish main occupations among hobbies or secondary occupations, (2) the difficulty to access representative sample among the different classes and (3) the presence of status words[17] which could bias the estimation [77]. But because our work don't try to predict SES but only to estimate types of occupation in order to link them with the sharing behaviours we don't care about representativity and we can attribute to each user more than one profession and status.

The main challenge for our work is then to be able to annotate as many

[17] For example : President, Head, Counselor...

users as possible with good precision, in our specific dataset. We tackle this challenge by replacing the automatic detection of professions by human annotated professions, focusing mainly on the most common declared occupations.

Here's our method :

1. Tokenize the description, removing punctuation and stop words[18]
2. Extract the most common words and bi-words
3. Recognize the words (or bi-words) referring to auto-declaration of an occupation (e.g. journalist, professor, city councilor...)
4. Link the declared occupation with the standard classifications (CSP2020, SOC2020, ISCO-08).
5. Highlight the professional classes and subclasses relevant for our analysis.

This technique not only allow us to obtain higher accuracy, because of human verification, but mainly allow us to identify more users by adapting the method to our specific dataset. We expected an accuracy over 70% [77], and a rapid check on our samples allow us to estimate over 80% accuracy.

Beacause our dataset is 80% french, we choose to analyze the professions collected using the french standard classification CSP2020. Table 2 we can see the main categories identified in our dataset. We can notice an over representation of highly educated positions (Cadres et professions intellectuelles supérieures) from information, politics and business fields. This bias has already been observed and it's mainly[19] due to the fact that professional from those categories has an higher probability of writing their professional titles on their description (among other things because they use their profile for professional purpose).

6.3 URLs informations

We consider finally the URL in our train set presented section 4.2. We tried to recognize known french medias among those URLs. We took as reference an extensive study that propose a classification of the french media ecosystem [25]. This study give us a data-set of 478 french medias. In this dataset we can find a classification of the medias obtained by identification of the groups resulting from Stochastic Block Model community detection [1] on the citation network among those medias. We presented the classification tab 3. In addition to that the dataset gives for each media several information on the type of media such as their outreach, their editorial structure or their origin.

We matched our collected URLs with medias when possible. Each media can be represented by more than one URL (e.g.: Le Monde can be lemonde.fr, blog.lemonde.fr...). In our analysis we will consider those URLs together as one media.

[18] in English and French

[19] Another reasons are the unequal population of Twitter users in general, and the collection method we used

Professional category	**Classification CSP2020** ***(French)***	**Users**	**Examples**
Information, arts and entertainment professions	Professions de l'information, de l'art et des spectacles (3500)	2315	journalist, producer...
Business, IT and administration professionals	Cadres et professions intermédiaires administratives, commerciales et techniques des entreprises (3700, 3800, 4600)	2160	developer, CEO...
Elected officers and political representatives	Cadres administratifs et techniques de la fonction publique (3300)	1192	mayor, minister...
Professors and higher scientific professions	Professeurs et professions scientifiques supérieures (3400)	636	professor, historian...
Other	Recognized professions in other CSP cathegories	530	nurse, architect...

Table 2: Main professional categories identified from users profile description. Professions related to information, politics and business are highly over represented.

Category	**Sub-categories**
Mainstream Media	Centre Hyper-centre
Opinion Journalism	Left Wing Right Wing
Periphery	Hyper-local (East) IT & Consumer Leisure and Health PQR & Mag
Counter-Informational Space	Alternative Health Identitarian Revolutionary Right

Table 3: Category of french medias proposed by [25]

We were able to recognize 440 different medias in our dataset, we call those media "recognized medias". In order to assess the coherence of our method and our data we can verify figure 4 that the number of interaction (i.e. sharing from Twitter users) for each media is linearly correlated with the number of citation that this media get from other medias.

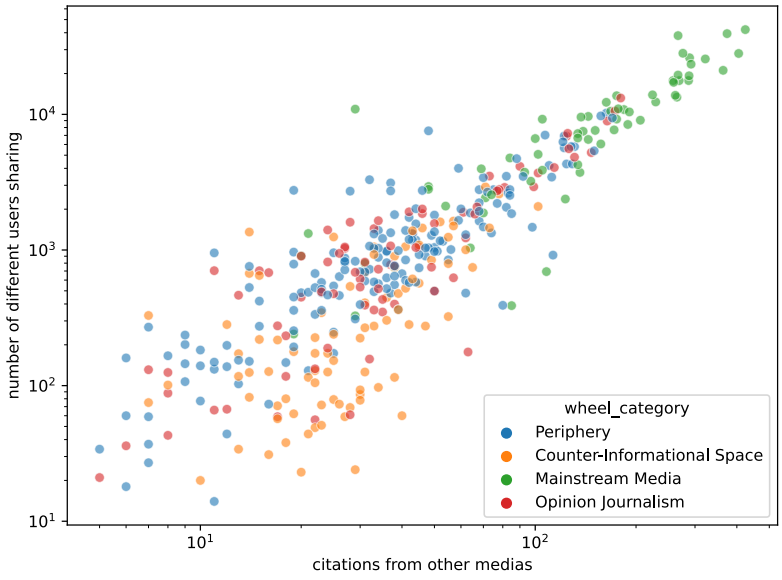
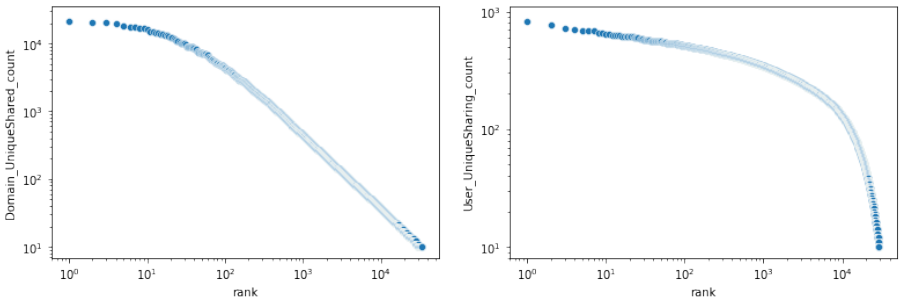


Figure 4: The number of different users sharing a media depending on the citations that this media get from the others.



(a) URLs ranked by number of interation (b) Users ranked by number of interaction

Figure 5: Distribution of the number of interaction per User and per URL

7 Input data analysis

We now have :

1. A user-URLs sharing dataset scraped from Twitter (section 4)
2. A working recommendation algorithm capable of recommending URLs to the users (section 5)
3. The embedding space used by the algorithm (section 5.5)
4. Information relative to the users : political attitudes, professions and languages (section 6)
5. Information relative to the URLs : the media from which they come from (section 6.3)

Our explanation method, presented section 8, will consist in finding relations between the embedding of the algorithm (3) and the user and URL external information (4-5). However, before that, we want to analyze our input data (1) under the light of our additional informations (4-5) in order to understand the structure of our input dataset. To do that, we fisrt study the size of the dataset section 7.1. We then compute a correspondence analysis (CA) of the sharing matrix (1). And we analyze the relation between the first 3 principal components (PC) of the CA with user's and URL's information (4-5) section 7.2.

7.1 Input data size

After the pre-processing phase 5.4 we obtained a dataset with 29.373 users, 32.639 URLs, and 3.277.738 interactions.

Looking at the distribution of the number of interaction for each URL on a log-log scale figure 5a, we can notice a constant gradient . This constant gradient is expected and typical of exponential distributions. In fact it is not absurd to

assume that the number of new sharings of a particular URL could depend linearly of the number of old sharings, giving then an exponential distribution.

This constant gradient does not appear on the log-log user interaction distribution figure 5b. We can in fact assume that real users[20] interactions don't scale exponentially because of time and resources limitations.

This quick analysis reassure us on the quality of our data.

7.2 Tendencies in the input data

In order to visualize and understand our input data we did a Correspondence Analysis CA of our Users-URLs interaction matrix. This way we obtained for each user and URL their coordinate in the 3 Principal Component (PC1, PC2, PC3) of the CA.

7.2.1 Medias coordinates in the CA embedding

Because each recognized media is represented by more than one URL we estimate the principal components of a media in the Correspondance Analysis by taking the weighted[21] average of the URLs corresponding to this media.

To assess the coherence of those principal components we looked at the weighted standard deviation of the coordinates of each URL belonging to the same media and we compare it to the weighted standard deviation of random media URLs of the same size. This comparison is showed figure 6 were we can see that the standard deviation of random group of URLs (red) is higher than the standard deviation of medias (blue), thus showing the coherence of those coordinate in analyzing media.

7.2.2 Interpretation of the principal components

We measure the distribution of each category of media, profession and language into the PCs, and compared it to the distribution of average users. We present and highlight here only the most relevant results. We argue after the analysis that the first 2 principal components represent mainly subject distinction separating specialized discussions (IT and business for PC1 and mainstream news for PC2), while the third dimension is mainly about geographical separation of the users.

Figure 7 we can see the distribution of each category of media on the 2 first principal components of the correspondence analysis. We can identify 2 zones of interest. The Information Technologies (IT) & consumers medias (e.g.: Businessinsider.fr, Canard PC, 01Net.com) occupies alone the area of positive PC1 and positive PC2. On the opposite side (negative PC1 and PC2) we can identify an area characterize by news media, in particular Hyper-center medias (which represent the mainstream french news medias such : BFMTV, Europe1.fr, Franceinter.fr).

[20] Not bots

[21] weighted on the number of interaction of each URL

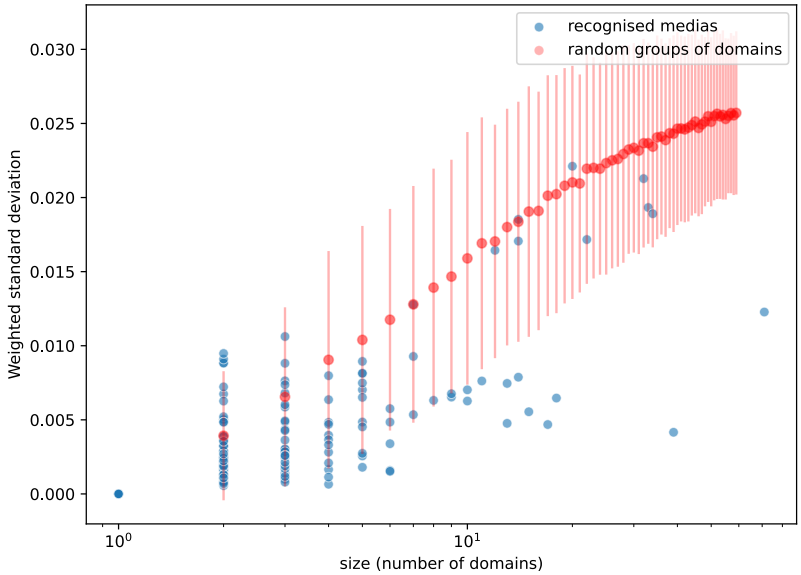
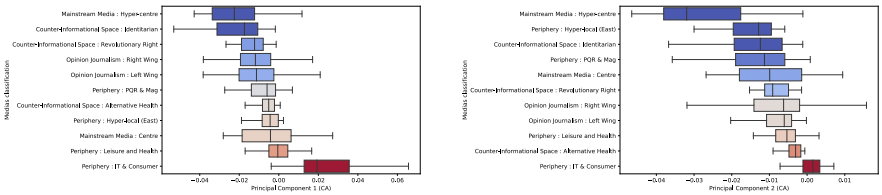


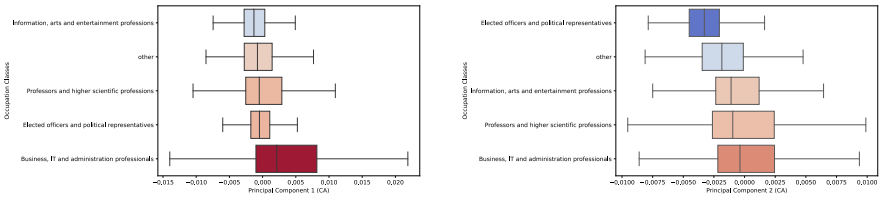
Figure 6: Comparison between the standard deviation of URLs belonging to the same media respect to random URLs groups



(a) **IT & Consumers medias** has an expected PC1 significantly higher than the average

(b) **news medias** (in particular mainstream news media) has an expected PC2 significantly lower than the average

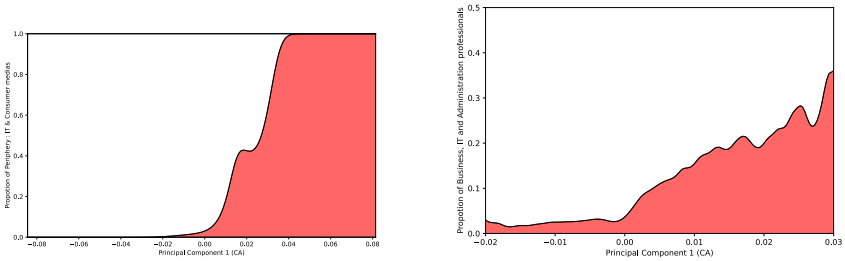
Figure 7: The distribution of each category of media on the 2 first principal components (PC1-PC2). The color depend on the median position of each category.



(a) As expected Business, IT and administration professionals tend to have positive PC1 (like IT and Consumers medias)

(b) As expected political professionals tend to have negative PC2 (like news medias)

Figure 8: The distribution of each professional category of users on the 2 first principal components (PC1-PC2). The color depend on the median position of each category.



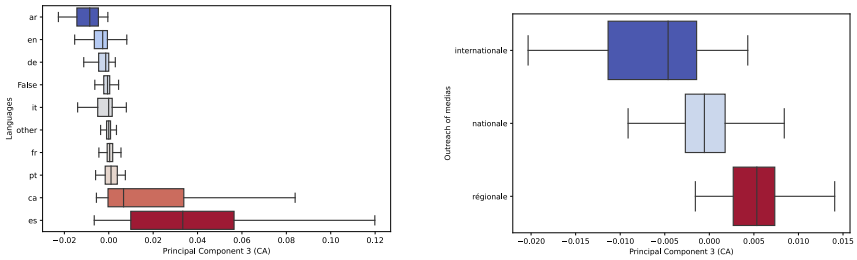
(a) Proportion of IT & consumer medias along the PC1.

(b) Proportion of business, IT and administration professionals along the PC1.

Figure 9: We estimate proportions by kernel density estimate (KDE), and we show that high PC1 users and media has an high probability to be business and IT related.

This distinction is confirmed by the profession analysis, figure 8 we can see the distribution of the professional categories on the principal components (PC). We notice, as expected, that the users classified as business, IT and administration professionals are indeed in a similar area than the IT & consumer medias, and that the political representatives professionals tend to be near to the mainstream news medias.

We can verify our analysis by plotting the proportion of IT & consumer medias and business, IT and administration professionals along the first principal component PC1 figure 9. This figure allow us to understand that : not only IT and business related medias and professional has a positive expected PC1, but also high PC1 users and medias has are probably IT and business related. It is then possible to characterise the principal component 1 as IT and business related.



(a) Distribution of user language along the PC3.

(b) Distribution of the geographical outreach of medias along the PC3.

Figure 10: The study the distribution of language and geographical related factors on the 3rd principal components (PC3). The color indicate the median shit of each distribution.

The third dimension (PC3) show little statistical correlations whit professions, or media categories but instead with language and geographical origins. In particular we can see figure 10a that the high PC3 area is dominated by spanish and catalan accounts and low PC3 by African accounts[22]. To confirm this hypothesis we can notice figure 10b that the geographical outreach of medias[23] has, in fact, an impact on the expected PC3 of those medias.

7.2.3 Political tendencies

We find political attitudes to be low explanatory factors for the first components of the CA of the input data.

7.3 Result of the factor analysis

We conclude the input data analysis by identifying 2 major factors in the sharing behaviors of users :

- The field of interest : separating business & IT related fields with political fields.
- The language of the users.

Those results are coherent with precedent research.

[22]Some qualitative analysis show also a great number of french speaking accounts from french speaking African countries like Congo

[23]i.e. the targeted audience

8 Political explanation

We developed section 5 a recommendation algorithm by matrix factorization able to recommend new URL to Twitter users from the URLs they shared. This algorithm worked by building an embedding of $k = 12$ dimensions with both the users and the URLs. We call those dimensions *latent dimensions learned by the algorithm*. In this chapter we are gonna propose an explanation method to understand these latent dimension. In particular we are gonna focus on political explanation, trying to understand if a relation exist between latent features and political attitudes.

In section 8.1 we present the state-of-the-art in algorithm interpretation and explanation, with a specific focus on NMF explanation techniques. In section 8.2 we present our work explaining our recommendation algorithm.

8.1 Interpretation and explanation state of the art

Before diving in our method it is important to understand the state of art of algorithmic explanation in order to situate our selves and to understand our design choices. There is a growing literature on explaining and interpreting algorithms. However, because the literature is still young, the methods and the terminologies are still unclear, we took the main survey studies as reference to locate our work in the state of art [89, 58, 90].

First we need to distinguish :

- **Interpretable model** : model that can itself provide humanly comprehensible interpretations (e.g. decision tree models, linear regressions, Bayesian model...).
- **Explanation method** : method that can make sense of a learned model after the training. Also called passive or *post-hoc* explanation methods.

In our work we want to build a *post hoc* explanation method (i.e. a method that don't try to change the model into an interpretable one but to explain the results).

Other important distinctions presented in literature are :

- **The level of explanation** : are we trying to get some formal rules explaining how the model works? Or are we trying to attribute some impact to specific input features?
- **The target of explanation** : do we explain only one result ? a group of result ? Or the whole model ?
- **The model specificity of the explanation method** : is the method specific to a determinate model, or can it be applied to a large family of models ?

We can see figure 11 a summary of some of those distinctions.

Dimension 1 — Passive vs. Active Approaches	
Passive	Post hoc explain trained neural networks
Active	Actively change the network architecture or training process for better interpretability

Dimension 2 — Type of Explanations (in the order of increasing explanatory power)	
To explain a prediction/class by	
Examples	Provide example(s) which may be considered similar or as prototype(s)
Attribution	Assign credit (or blame) to the input features (e.g. feature importance, saliency masks)
Hidden semantics	Make sense of certain hidden neurons/layers
Rules	Extract logic rules (e.g. decision trees, rule sets and other rule formats)

Dimension 3 — Local vs. Global Interpretability (in terms of the input space)	
Local	Explain network's *predictions on individual samples* (e.g. a saliency mask for an input image)
Semi-local	In between, for example, explain a group of similar inputs together
Global	Explain the network *as a whole* (e.g. a set of rules/a decision tree)

Figure 11: The explanation method taxonomy proposed by [90]

According to the classification proposed our approach is:

- a passive post-hoc explanation (we don't act on the model)
- an attribution of hidden semantics. I.e. it assign impact of some -*implicit*- input features[24] (attribution) to an hidden layer of the model[25] (hidden semantics)
- semi-local (concern a subset of users and latent features)
- semi-model agnostic (can be applied to all recommendation models with embedding layers)

8.1.1 Explanation for matrix factorization models

We are trying to build an explanation for a Non-Negative Matrix Factorization algorithm. This task has already been tackled by the past. We can first cite several attempts to make the matrix factorization interpretable : building a bayesian NMF model were the matrix factors can be interpret as probability [39], powering the model with user and item aspect information [40] or leveraging overlapping clusters of users and items to improve performance and explain the recommendation [83]. Beside that, some post-hoc explanation has also been developed : for example by explaining recommendation with nearest neighbors analysis on the latent space [2].

An interesting work to cite is the one of Datta et al. [27]. This work (applied to film recommendation) has in common with us the fact of explaining by attribution of *implicit* input variables. In fact we try to explain recommendation by

[24]The political attitudes
[25]The latent embedding

attributing impact of political and socio-demographic features, although these features are not really part of the input features of the algorithm, they only implicitly impact the input features[26] as we see in section 7.2.

However our method bring novelty in 2 points : (1) it leverage political attitudes, (2) it doesn't explain the result of recommendation but directly the semantic of the latent embedding.

8.2 Explanation of our algorithm

Now let's dive into our explanation method. We try to explain the 12 dimensional latent embedding resulting from our recommendation algorithm. This embedding dimensions are expected to be insightful of the user preference because, as we said section 5, the scalar product of the coordinate of an user with the one of an object should give us the predicted rating of this user for this item.

8.2.1 Attribution methods

For each user we then obtain : his political attitude section 6.1, his profession and language section 6.2 and his algorithmic embedding section 5. In this section we will consider those variable as random variables and study the statistical relationships between those variables.

We will call Φ_{LR} and Φ_{AE} respectively the Left-Right attitude and the Anti-Elite salience attitude.

We will call $L_0, ..., L_{11}$ the latent dimensions learned by the algorithm.

Each user will be considered as a drawing of those random variables.

The idea of attribution methods is to quantify how a change in the input features (here Φ_{LR} and Φ_{AE}) will impact a change in the result of the algorithm (here $L_0, ..., L_{11}$).

For a model m and a feature vector $\mathbf{x}$, we can define the impact attributed to a feature x_i as :

$$\text{Attr}(x_i) := \mathbb{E}_{x_i}[m(\mathbf{x}) - m(\mathbf{x}_{-x_i})]$$

Where $\mathbf{x}_{-x_i}$ is the feature vector without the feature i.

Unfortunately in the case of external features we cannot really estimate $m(\mathbf{x}_{-x_i})$. We aim to adapt the formula to our case, and propose a new formal attribution method.

8.2.2 Statistical correlations

For now we don't know how to define attribution in case of implicit input features in our case. Our main goal here is then to highlight the conditional relations between attitude and latent features, in order to understand what kind of relation there is and be able to propose a suitable attribution formula.

[26] we then call them implicit input features

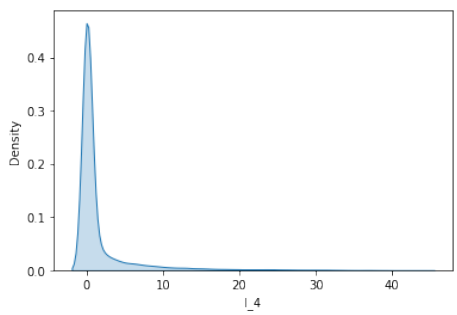


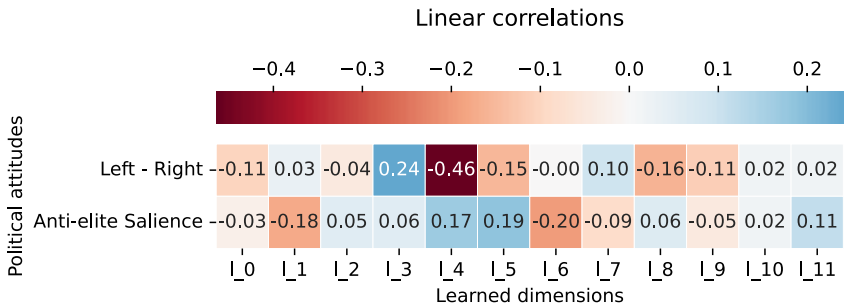
Figure 12: Example of distribution of a latent dimension.

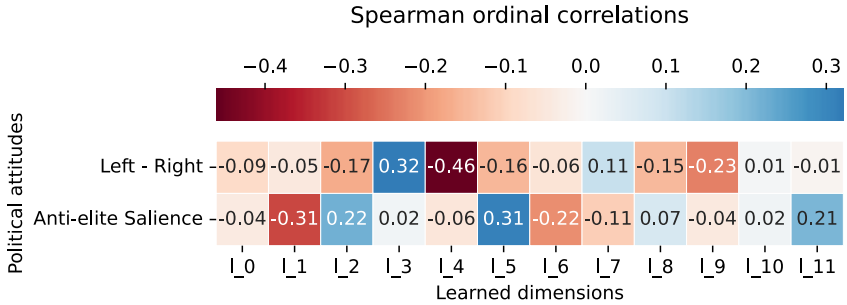
To get ourselves an informal idea of what the NMF algorithm is doing in our case of application we can consider that the algorithm is trying to recognize *sharing behaviors*. Each dimension of the latent embedding will represent a certain *sharing behavior* associated to a certain group of item and a certain group of users, for each user the learned latent vector represent the degrees of belonging to each group (or sharing behavior). The role of the algorithm is to identify which groups and which degrees of belonging will maximise the likelyhood of the observed interaction (train set). In particular we can notice that each group (hence each latent dimension) contain only a little part of all the users. For this reason, if we plot the distribution of the users on each latent dimension (see figure 12) we expect a very fast decreasing distribution.

As presented section 8.2.1, we are trying to highlight the statistical relations between user features (in particular ϕ_{LR} and ϕ_{AE}) and learned dimensions $(L_0, ..., L_{11})$.

We first need to identify the latent dimensions of interest. We find them measuring the correlation between Φ_{LR}, Φ_{AE} and each latent dimension $L_0, ..., L_{11}$. We compute for each couple the linear correlation (Pearson), the ordinal correlation (Spearman and Kendall-tau) and the mutual information. We can see figure 13 the result of this correlation measure. Firstly we can see is that the latent learned features contain in fact political information. Secondly we can highlight some dimensions (for example L_3 and L_4) to have particularly high correlations ($p - value < 10^{-80}$)[27] with political attitudes (specifically with the Left-Right attitude).

In the next section we want study the specific nature of this relation. We will study what L_3, L_4 tell us about the political attitudes of users Φ_{LR}, Φ_{AE} section 8.2.3 and we study how Φ_{LR}, Φ_{AE} impact the expected L_3, L_4 section 8.2.4.





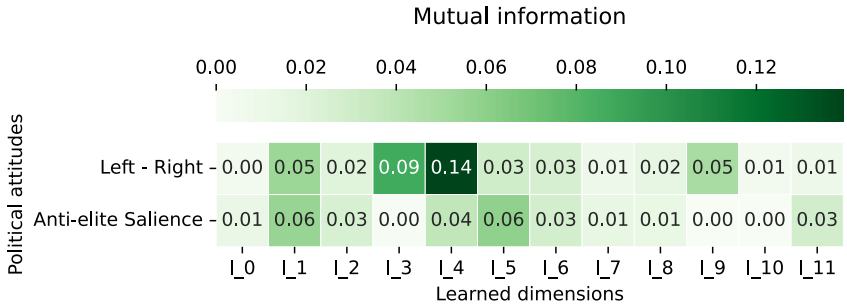


Figure 13: Relation between political attitudes and learned latent dimensions. We remember that this correlation are measured on samples of approximately 30.000 users, giving then an high statistical significance to the result.

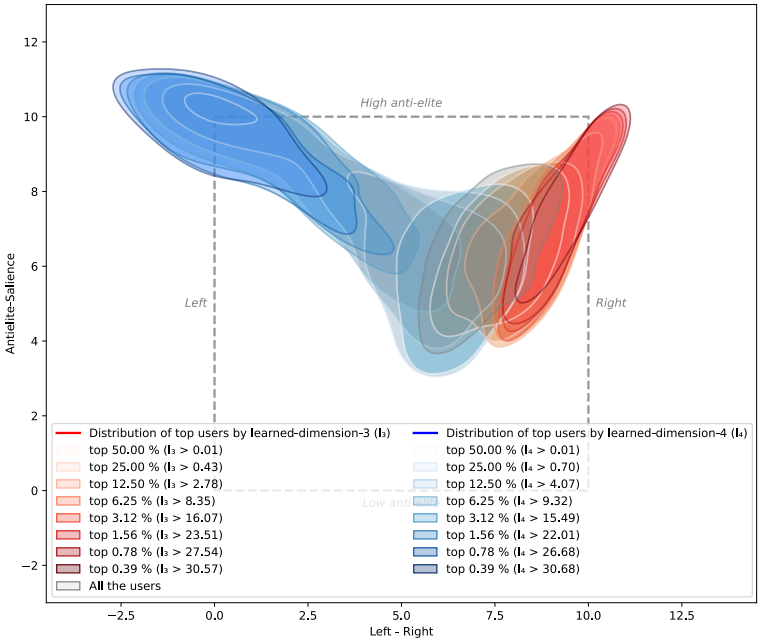


Figure 14: Conditional distribution of users political attitudes depending on the learned dimensions 3 (l_3 in red) and 4 (l_4 in blue). Political attitudes distribution of the user with high l_3 and l_4 are significantly different than the distribution of average users

8.2.3 Expected political attitudes depending on latent dimensions

We study the conditional joint distribution of the political attitudes Φ_{LR} and Φ_{AE} depending on the learned dimensions L_3 and L_4. Formally :

$$f_{\Phi_{LR},\Phi_{AE}|L_3,L_4}(\phi_{LR},\phi_{AE})$$

In particular in figure 14 we show that the political attitudes distribution of the user with high l_3 and l_4 are significantly different than the distribution of average users.

Let l_3^* and l_4^* be 2 fixed thresholds respectively for L_3 and L_4 and $p^* \in [0,1]$ a probability threshold.

For a given latent dimension j with a given threshold l_j^*, we define the *high density area* of j, $D^{(j)}_{l_j^*,p^*}$ as the political attitudes area where the density of users with $L_j > l_j^*$ is higher than p^*.

Formally for L_3 and L_4 we get :

$$\begin{aligned} D^{(3)}_{l_3^*,p^*} &= \{(\phi_{LR},\phi_{AE}) \text{ s.t. } f_{\Phi_{LR},\Phi_{AE}|L_3>l_3^*}(\phi_{LR},\phi_{AE}) \geq p^*\} \\ D^{(4)}_{l_4^*,p^*} &= \{(\phi_{LR},\phi_{AE}) \text{ s.t. } f_{\Phi_{LR},\Phi_{AE}|L_4>l_4^*}(\phi_{LR},\phi_{AE}) \geq p^*\} \end{aligned} \tag{1}$$

In figure 14 we show $D^{(3)}_{l_3^*,p^*}$ and $D^{(4)}_{l_4^*,p^*}$ for different thresholds l_3^* and l_4^* with $p^* = 0.5$.[28] What we observe there is that users with high l_3 have an high probability to have right wing and anti-elite attitudes and users with high l_4 have an high probability to have left wing and anti-elite attitudes.

The model is then actually inferring political attitudes.

8.2.4 Expected latent dimensions depending on political attitudes

We estimate the expected values of L_3 and L_4 conditionally to the political attitudes of the users (Φ_{LR} and Φ_{AE}). For a given left - right attitude ϕ_{LR} and a given anti-elite attitude ϕ_{AE} we try to predict the expected value of L_3 :

$$\overline{l_3}(\phi_{LR},\phi_{AE}) = \mathbb{E}[L_3|\Phi_{LR}=\phi_{LR},\Phi_{AE}=\phi_{AE}] \tag{2}$$

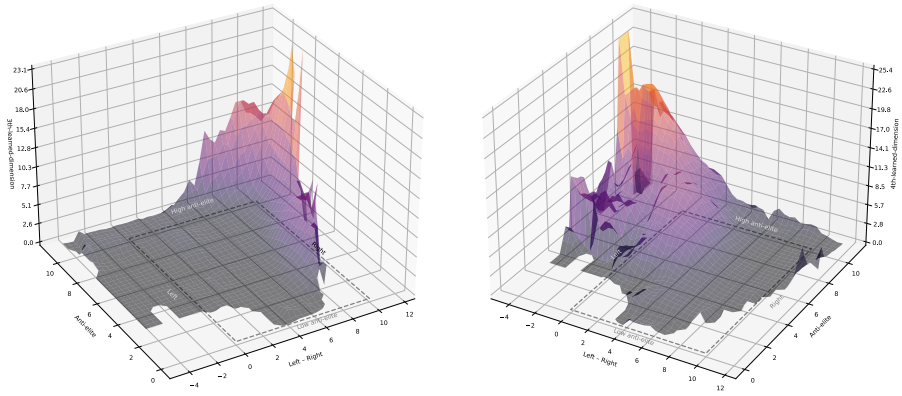
We estimate it by computing the "local" (in terms of attitudes) average of L_3 (respectively L_4).

We introduce the concept of "locality" of (ϕ_{LR},ϕ_{AE}) by defining the *attitudinal neighborhood* of (ϕ_{LR},ϕ_{AE}) as :

$$\begin{aligned} \mathcal{U}_{\phi_{LR},\phi_{AE},\Delta\phi_{LR},\Delta\phi_{AE}} := \{u \in \mathcal{U} \text{ s.t. } & \phi^u_{LR} \in [\phi_{LR}-\Delta\phi_{LR},\phi_{LR}+\Delta\phi_{LR}] \\ & \text{and } \phi^u_{AE} \in [\phi_{AE}-\Delta\phi_{AE},\phi_{AE}+\Delta\phi_{AE}]\} \end{aligned} \tag{3}$$

[27] The high p-value is due to the size of the dataset

[28] $p^* = 0.5$ has been chosen for visualization reasons, we can see the same effect for different values of p^*



(a) The expected value of the 3th learned dimension (l_3) depending on the political attitudes of the users. The expected value of l_3 is higher for right wing users.

(b) The expected value of the 4th learned dimension (l_4) depending on the political attitudes of the users. The expected value of l_4 is higher for left wing users.

Figure 15: Expected value of l_3 and l_4 depending on political attitudes

In our case we use $\Delta\phi_{LR} \approx 0.9$ and $\Delta\phi_{AE} \approx 0.7$.[29]

We estimate $\overline{l_3}$ by measuring the average l_3 of the attitudinal neighborhood :

$$\overline{l_3} \approx \hat{l_3} = \frac{\sum_{u \in \mathcal{U}_{\phi_{LR},\phi_{AE}}} (l_3^u)}{|\mathcal{U}_{\phi_{LR},\phi_{AE}}|} \tag{4}$$

Figure 15a we can see the expected $\hat{l_3}$ depending on ϕ_{LR} and ϕ_{AE}. And figure 15b the expected $\hat{l_4}$.

This result suggest that not only the model is inferring the political attitudes of some user but also that the model is leveraging these attitudes giving them an impact on the position in the embedding space.

8.3 Conclusion of the analysis

We studied in detail the relation between Φ_{LR}, Φ_{AE} and L_3, L_4 in section 8.2.3 and section 8.2.4.

Before jumping to conclusion is important to acknowledge that those latent dimensions do not learned the whole left-right or anti-elite salience dimensions. In fact we do not see a complete linear relations between the 2 variables, e.g.: not all the left wing users will be embed by the algorithm as high l_4.

[29] In particular we choose $\Delta\phi_{LR} = \frac{1}{16}(max_{u \in \mathcal{U}}(\phi_{LR}^u) - min_{u \in \mathcal{U}}(\phi_{LR}^u))$ respectively ϕ_{AE} for $\Delta\phi_{AE}$

However we argue that the latent dimensions 3 and 4 learned sharing behaviors that are specific to the right or the left side of the political spectrum. In fact high l_4 medias are very specific left wing medias, and users embed as high l_4 are nearly all left wing users.

These result will be the base for the development of our formal explanation method.

8.4 Further works

Obviously, because this section is a work in progress, most of the "further work" will be proposed here.

8.4.1 Sociodemographic explanation

The first work (now in progress) is to remake the same explanation analysis with the socio-demographic datas. This work has 2 goals :

- Extend our explanation method to socio-demographic data, and in particular to cathegorical random variables.
- Look for potential confounding factor in our political explanation.

The first result seems to confirm that some latent dimensions are in fact learning only political attitudes and not professions or languages. However more work need to be done in this direction.

8.4.2 Formal explanation

As we describe it previously our explanation method hasn't been formalize and generalize to all the recommender embedding attributions with implicit input features. In the next months we hope to be able to formalize the mathematical tools we use in order to be able to propose a proper explanation method.

We will then be able to scrutinize our method to the standards of explanation techniques [80].

8.4.3 Sociological explanation techniques

During the decades different forms of algorithm explanation has been proposed. Social sciences (and in particular sociology) aren't missing from those propositions.

Back in 1985 Woolgar has already introduced the idea of applying sociological methods to study machine behaviors, and in particular to study their impact on social phenomenas [87]. This idea has since been retaken and evolved in different form, one of them being the concept of machine habitus [4], which try to apply the habitus theory [16] on agent entities like AI systems.

One advantage of leveraging social theories to explain recommender systems behaviors could be the possibility to account for the environment (public digital

space, algorithm governance, emerging technologies...) as part of the explanation and not only as context.

We aim in this thesis to situate ourselves on a truly interdisciplinary position. And, while the possibilities are still vague now, we have the long term goal to propose at least an idea of socio-technical framework to understand recommendation systems in their environment.

9 Deep Learning recommender systems

One of the long therm goal of this thesis is to be able to propose a working explanation technique for recent Deep Learning powered recommendation algorithms.

In parallel to the work on collaborative filtering I have been working on graph neural network algorithms for link recommendation. This kind of model are becoming most and most common in the recommendation algorithms literature and are often cited as part of real world recommender systems [88]. I studied the litterature, and I implemented and started to train a few GNN models. However because of the time consumption of the task I have put the training of this large model in stand-by from October 2022.

For this reason I won't present this work here. I will only let some chapters of my internship report as appendix A if some reader want to research the question.

This being said the GNN recommender system is still on our to do list. We plan to retake the model created and to apply it not to a user-user graph (as presented in the appendix) but to the same user-item graph we already have. We hope that this will allow us to make direct comparisons between the GNN embedding and the NMF one.

10 Pratical operation

The thesis project started may 2022 with a 5 month internship. During the internship I worked on :

- Writing the thesis project
- Gathering and cleaning the Twitter data from the medialab
- Developing 3 recommendation algorithms : 2 NMF item recommender and 1 GNN link recommender

The reference documentation for this work is my internship report. Main results where presented during the CIVICA research conference days 2022.

Starting September 2022, I extracted the embedding from the best working NMF algorithm and start to analyze it. I also went more deeply in the litterature arround explicability and social media studies. Until february 2023 they where frequently new (and better performing) versions of the algorithm, and new tools to study and plot the statistical relations.

From march 2023 we had the goal to submit a paper for the 17th ACM Conference on Recommender Systems. I missed the paper submission deadline, so we submitted to the last breaking results early July. Unfortunately the paper was rejected.

From march 2023 to July 2023 we also worked on the profession, language, and media cathegory estimation. I completed the estimations and did the analysis in June 2023.

February 2023 a poster was accepted to IC2S2 2023 it was presented in July 2023.

August 2023 we submitted to ASONAM 2023.

September 2023 a poster submission was accepted for the CIVICA research conference days 2023.

11 Conclusion

Research conclusions has already been presented section 8.3. And there is no much to conclude here because the PhD work is far from concluded. However because it feels harsh to let the reader without a conclusion I put this chapter anyway. To reward the reader that came this far, I will conclude positively saying that : I'm satisfied of the work done for now, and optimistic (and curious) about the future !

If there are any questions about elements in this document, they can be addressed at : tim.faverjon@sciencespo.fr .

A Appendix : Documentation about Graph Neural Networks

A.1 Contexte scientifique

Commençons par quelques conceptions générales et un peu d'histoire autour des *Graph Neural Networks* (GNN) ou Réseaux de Neurones sur Graphes.

Les GNNs ne sont pas spécifiques aux tâches de recommandation. Ils sont simplement spécifiques des graphes et sont donc utilisés dans tous les domaines où nous pouvons représenter les données sous forme de graphes. Ils sont a fortiori typiques de l'étude des réseaux des représentations géographiques et, plus récemment, ont commencé à être largement employés en biologie.

L'idée de base des GNN est de construire des convolutions entre couches d'un réseau de neurones se basant sur la proximité des éléments sur le graphe (de la même façon dont les réseaux de convolutions se basent sur la proximité géographique des pixels entre eux). Concrètement, cela se traduit souvent mathématiquement par l'utilisation de couches dont au moins une dimension correspond au nombre de nœuds et à multiplier la couche en question par la matrice d'adjacence du graphe sur cette dimension. Dans la section A.1.1 je présente plus en détails la technique en introduisant quelques formalisations mathématiques.

Un des premiers travaux fondateurs introduisant l'idée de GNN a été celui de Perozzi et al. en 2014 [69]. Dans ce papier, Perozzi résout un problème de classification de nœuds sur un graphe en introduisant l'idée de marche profonde (Deep Walk). L'idée est d'enrichir le graphe au-delà des simples liens de premier degré avec des liens (à poids plus faibles) issus de marches aléatoires sur le graphe. L'idée principale introduite ici est celle de propager de l'information à travers les liens du graphe. C'est à partir de cette idée que Kipf et al. [49] proposent, en 2017, un des premiers modèles de *Graph Convolution Network* (GCN), toujours pour une tâche de classification de nœuds (cette fois-ci semi-supervisée). Ils présentent alors la méthode comme une approximation de la convolution spectrale sur le graphe.

Depuis beaucoup de variantes et d'évolutions ont succédé au modèle de base de GCN. En particulier, on peut considérer les modèles d'attention, *Graph Attention Network* (GAT) [82], ou encore les modèles dits de *message passing* et ceux de GraphSAGE. Le rôle de ces modèles reste cependant le même : faire propager l'information d'un nœud à ses voisins de façon à ce qu'à la fin du processus, le vecteur associé à chaque nœud contienne des informations relevant de son voisinage dans le graphe.

Dû à la facilité et à l'habitude de représenter les informations liées aux systèmes de recommandation sous forme de graphe, les GNN ont vite été employés pour des tâches de recommandation. L'avantage global de ces méthodes est qu'elles peuvent intégrer à la fois des données objet, des données utilisateur, des données sociales et des données d'interactions utilisateur-objet. Elles peu-

vent donc englober toutes les catégories des systèmes de recommandation en un seul modèle.

Il existe beaucoup de structures possibles pour construire des algorithmes de recommandation avec des outils d'apprentissage profond[88]. Nous pouvons cependant distinguer trois phases communes aux algorithmes de recommandations par GNN :

1. Structuration des données : nous passons de données brutes à des données sous forme de graphe (ex.: graphe social des utilisateurs, graphe d'interactions utilisateur-objet, graphe d'actions des utilisateurs...).

2. Apprentissage sur graphe : à partir du graphe, nous appliquons un GNN pour redéfinir ou attribuer un vecteur à chaque utilisateur et/ou chaque lien (ex.: GCN, GAT, SAGE...).

3. Couche prédictive : à partir du graphe entraîné, nous construisons le prédicteur adapté à nos besoins (ex.: Prédicteur d'apprentissage classique comme des régressions logistiques, classifieur d'apprentissage profond...).

Dans ce travail, nous allons nous intéresser à la problématique spécifique de **prédiction de liens** sur graphe. Dans ce paradigme, nous considérons uniquement les utilisateurs, nous partons d'un graphe social entre utilisateurs (potentiellement enrichi par des informations personnelles pour chaque nœud), pour lequel nous ignorons certains liens et nous cherchons à prédire les liens manquants. Cette problématique étant, évidemment, pensée pour la recommandation d'amis sur les réseaux sociaux.

Le problème de prédiction de liens a d'abord été formalisé mathématiquement dans le cadre des *network sciences* [56]. Au fil des années, les techniques de résolution passent graduellement de formalisation mathématique et physique pour proposer toujours plus de modèle d'apprentissage profond, ces modèles ayant l'avantage de pouvoir facilement intégré les caractéristiques des utilisateurs à leur calculs [36, 85].

N'ayant pas fait de réelles revues de l'état de l'art sur le sujet nous avons choisi simplement de nous baser sur un benchmark déjà existant : le Open Graph Benchmark. Notre démarche est présentée dans la section suivante.

A.1.1 Le modèle

L'architecture complète

On peut voir figure 16 l'architecture globale du modèle que nous utilisons, ce modèle a été développé par Wang et al. [86]. Il introduit en particulier la notion d'apprentissage par paire (Pairwise Learning), concept que nous expliquons plus loin. On peut voir, comme expliqué dans la section A.1, que nous avons trois phases : le pré-traitement et l'échantillonnage (gauche), l'encodeur GNN (centre) et le prédicteur (droite).

Dans la cadre de ce stage, nous n'utilisons pas les *Node-Pair Neighborhood Encoder*, mais nous pensons explorer la possibilité par la suite.

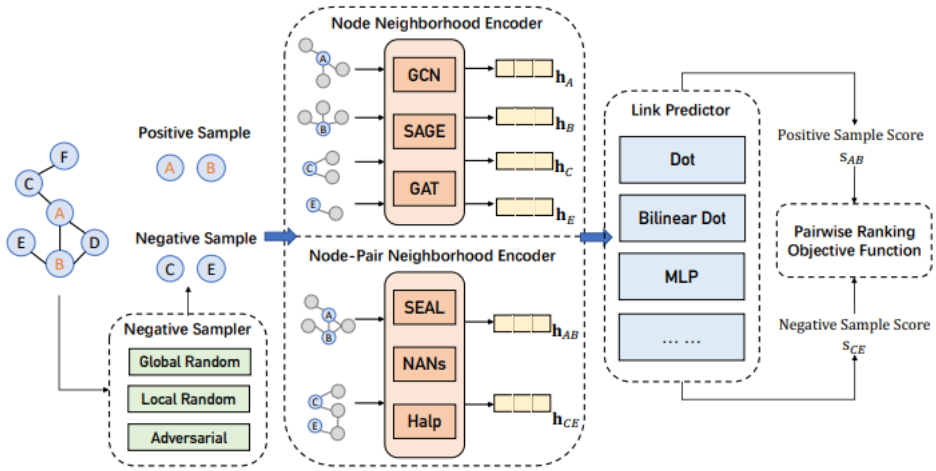


Figure 16: modèle de prédiction de liens PLNLP (issu de [86])

Les encodeurs par GNN :

Soit $\mathcal{G} = (\mathcal{U}, \mathcal{L})$ le graphe social décrivant les liens entre utilisateurs. On note A la matrice d'adjacence de ce graphe et, pour tout utilisateur $u \in \mathcal{U}$, on note $\mathcal{N}_u = \{v \in \mathcal{U} | (u, v) \in \mathcal{L}\}$ l'ensemble des voisins de u. On associe à chaque utilisateur un vecteur de caractéristiques : $\mathbf{x}_u, \quad \forall u \in \mathcal{U}$. L'idée des GNN est de construire pour chaque utilisateur u un vecteur $\mathbf{h}_u^{(final)}$ qui incorpore à la fois les informations issues de l'utilisateur que de son voisinage.

Compte tenu des données du graphe, l'idée principale des GNN est d'agréger de manière itérative les informations sur les caractéristiques des voisins et d'intégrer les informations agrégées avec la représentation actuelle du nœud central pendant le processus de propagation. Du point de vue de l'architecture du réseau, les GNN empilent plusieurs couches de propagation, qui consistent en des opérations d'agrégation et de mise à jour. La formulation de la propagation est :

$$\begin{aligned} \text{Agrégation:} \quad & \mathbf{n}_v^{(l)} = \text{Aggregator}_l \left(\left\{ \mathbf{h}_u^l, \forall u \in \mathcal{N}_v \right\} \right), \\ \text{Mise à jour:} \quad & \mathbf{h}_v^{(l+1)} = \text{Updater}_l \left(\mathbf{h}_v^{(l)}, \mathbf{n}_v^{(l)} \right), \end{aligned} \tag{5}$$

Où $\mathbf{h}_u^l$ désigne la représentation du nœud u sur la couche l, et Aggregator_l et Updater_l représentent la fonction d'opération d'agrégation et de mise à jour sur la l^{ieme} couche, respectivement.

Il existe plusieurs formes de GNN que l'on peut voir représentées figure 17. Ces formes se différencient en fonction de la manière dont elles agrègent l'information et mettent à jour les nœuds. Elles sont ici ordonnées de la moins complexe à la plus complexe.

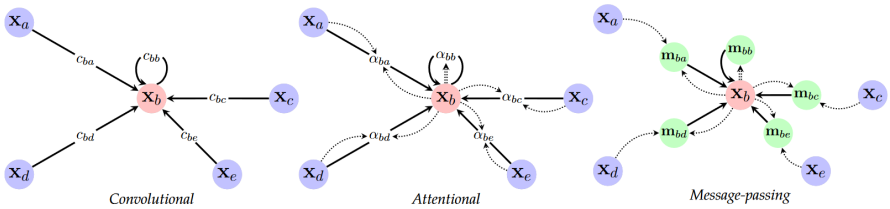


Figure 17: Différents types de GNN (issus du manuel [18])

- **GCN :** [49]

  Les *Graph Convolution Networks* sont en fait une approximation au premier ordre de la décomposition propre du Laplacien du graphe. Dans ce contexte, chaque lien a un poids fixe, déterminé à l'avance par la matrice d'adjacence, figure 17 on a $c_{uv} = A_{uv}$.

  Concrètement, on a :

$$\begin{aligned} \text{Aggregation:} \quad & \mathbf{n}_v^{(l)} = \sum_{j \in \mathcal{N}_v} d_{vv}^{-\frac{1}{2}} \tilde{a_{vu}} d_{uu}^{-\frac{1}{2}} \mathbf{h}_u^{(l)}, \\ \text{Update:} \quad & \mathbf{h}_v^{(l+1)} = \delta\left(W^{(l)} \mathbf{n}_v^{(l)}\right), \end{aligned} \tag{6}$$

  Avec $\delta(.)$ une fonction d'activation non-linéaire (ReLU...), $W^{(l)}$ la matrice des paramètres de transformation apprise pour la couche l, $\tilde{a_{uv}} = [I_n + A]_{uv}$ le poids du lien de u vers v (avec $\tilde{a_{uu}} = 1$) et $d_{uu} = \sum_v \tilde{a_{uv}}$.

- **GraphSAGE :** [35]

  Similaire aux autres méthodes, le GraphSAGE se distingue par la concaténation de l'agrégation $\mathbf{n}_v^{(l)}$ et du vecteur utilisateur $\mathbf{h}_v^{(l)}$ dans la phase de mise à jour. On obtient donc :

$$\begin{aligned} \text{Aggregation:} \quad & \mathbf{n}_v^{(l)} = \text{Aggregator}_l\left(\left\{\mathbf{h}_u^l, \forall u \in \mathcal{N}_v\right\}\right), \\ \text{Update :} \quad & \mathbf{h}_v^{(l+1)} = \delta\left(W^{(l)} \cdot \left[\mathbf{h}_v^{(l)} \oplus \mathbf{n}_v^{(l)}\right]\right), \end{aligned} \tag{7}$$

  Où Aggregator_l est la fonction d'agrégation de la couche l (moyenne, somme, max-pooling...), et, comme plus haut, $\delta(.)$ est une fonction d'activation non-linéaire (ReLU...) et $W^{(l)}$ la matrice des paramètres de transformation apprise pour la couche l.

- **GAT :** [82]

  Les *Graph Attention Networks* (GAT) utilisent le même principe que les GCN, mais partent du principe que l'influence de chaque voisin n'est pas

identique et pas pré-établie par le graphe, et qu'elle peut donc être entraînée pour chaque tâche spécifique. On rajoute donc pour cette tâche une valeur d'attention pour chaque lien, calculée à partir des nœuds correspondants. On peut voir figure 17 (au centre) que les GAT sont un intermédiaire entre les GCN et les modèles dits de *Message-Passing*. En effet, dans ces derniers, l'entièreté du message agrégé est déterminée par le vecteur de départ et celui d'arrivée, dans les GCN, c'est uniquement le vecteur de départ qui est agrégé, tandis que dans les GAT nous avons l'attention qui est déterminée par l'arrivée et le départ, et le message qui est déterminé uniquement par le départ. En plus d'avoir prouvé leur efficacité sur le terrain, les GAT ont l'avantage de pouvoir différencier le poids de chaque lien sans augmenter de façon trop importante les paramètres du modèle et en évitant donc le sur-apprentissage.

Dans notre cas spécifique, nous verrons que ces réseaux ont également des propriétés particulières en termes d'explicabilité. En effet, on considère souvent les attentions comme une forme d'explication de l'impact de chaque lien dans les résultats finaux.

$$\begin{aligned} \text{Aggregation:} \quad & \mathbf{n}_v^{(l)} = \sum_{j \in \mathcal{N}_v} \alpha_{vj} \mathbf{h}_j^{(l)}, \alpha_{vj} = \frac{\exp\left(\text{Att}\left(\mathbf{h}_v^{(l)}, \mathbf{h}_j^{(l)}\right)\right)}{\sum_{k \in \mathcal{N}_v} \exp\left(\text{Att}\left(\mathbf{h}_v^{(l)}, \mathbf{h}_k^{(l)}\right)\right)}, \\ \text{Update :} \quad & \mathbf{h}_v^{(l+1)} = \delta\left(W^{(l)} \mathbf{n}_v^{(l)}\right), \end{aligned} \tag{8}$$

Où Att est une fonction d'attention. On a donc $\text{Att}(\mathbf{h}_v^{(l)}, \mathbf{h}_j^{(l)}) = a(W^{(l)}\mathbf{h}_v^{(l)}, W^{(l)}\mathbf{h}_j^{(l)}) \in \mathbb{R}$. On utilisera par la suite la formule d'attention classique :

$$\text{Att}(\mathbf{h}_v^{(l)}, \mathbf{h}_j^{(l)}) = \text{LeakyReLU}(\mathbf{a}^T[W^{(l)}\mathbf{h}_v^{(l)} \oplus W^{(l)}\mathbf{h}_j^{(l)}]) \tag{9}$$

Ici, $\mathbf{a}$ est un vecteur de paramètres appris par le modèle. Comme plus haut, $\delta(.)$ est une fonction d'activation non-linéaire (ReLU...) et $W^{(l)}$ la matrice des paramètres de transformation apprise pour la couche l.

Le prédicteur

Une fois que chaque utilisateur u est encodé par un vecteur $\mathbf{h}_u^{(l_{final})}$, où l_{final} représente le nombre total de couches de GNN utilisées, nous devons associer à chaque couple (u, v) une valeur associée à la probabilité que ce couple soit connecté $(u, v) \in \mathcal{L}$.

Les prédicteurs sont simplement des fonctions associant à deux vecteurs $\mathbf{h}_u^{(l_{final})}$ et $\mathbf{h}_v^{(l_{final})}$ un score $s_{uv} \in \mathbb{R}$. Nous utilisons deux prédicteurs :

- **Dot Predictor :** Le prédicteur le plus simple est bien sûr le produit scalaire.

$$s_{uv} = \mathbf{h}_u \cdot \mathbf{h}_v \tag{10}$$

  Nous utilisons ce prédicteur quand nous considérons le graphe non dirigé.

- **Bilinear Dot Predictor :** Quand nous considérons notre graphe comme un graphe dirigé, nous pouvons en revanche introduire une matrice de paramètres apprenables $W^{(pred)}$ et définir le score par :

$$s_{uv} = \mathbf{h}_u W^{(pred)} \mathbf{h}_v \tag{11}$$

Nous pourrions remarquer que les prédicteurs utilisés sont très simples, et nous demander : mais pourquoi donc ?

Bien que cela soit rarement explicité dans la littérature, une des raisons identifiées est d'éviter le sur-apprentissage. En effet, un prédicteur trop complexe et en mesure de sur-apprendre sur les données disponibles empêcherait les couches d'encodeur de s'entraîner proprement[30]. Pour nous, cette réalité apporte également un autre avantage : l'explicabilité. En effet, plus le prédicteur est simple plus nous pouvons considérer que les informations relatives aux liens des utilisateurs sont contenues dans l'espace des représentations entraînées $H^{(l_{final})}$. En particulier dans le cas du *Dot Predictor*, nous nous situons exactement dans le cas de la factorisation matricielle présentée section 5. Si nous considérons un problème où nous ne connaissons aucune caractéristique des utilisateurs alors, à une fonction monotone près appliquée à s_{uv}, nous serions en fait en train de résoudre un problème de factorisation matricielle.

De manière générale, on remarquera que n'importe quel classifieur bi-classe peut faire office de prédicteur.

Hyperparamètres du modèle :

- `encoder` : l'encodeur GNN choisi pour le modèle (voir section A.1.1)
  - `GCN` : Graph Convolution Network
  - `SAGE` : GraphSAGE
  - `GAT` : Graph Attention Network
- `gnn_num_layers` : f_{final} le nombre de couches de GNN utilisées.
- `gnn_hidden_channels` : la dimension des représentations vectorielles $\mathbf{h}_u^{(l)}$ (ce qui impacte également la dimension de la matrice de paramètres $W^{(l)}$ en fonction du modèle).

[30] On remarquera que la littérature se contente parfois également d'un simple "ça marche très bien"

- `emb_hidden_channels` : le nombre de caractéristiques initiales par utilisateur, donc la dimension du vecteur $\mathbf{x}_u$.
- `predictor` : le prédicteur utilisé (voir section A.1.1)
  - `DOT` : Produit vectoriel
  - `BIL` : Produit matriciel bilinéaire

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